

FL Simulation Results

File: test_A9

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8747
Clean + DP Protection Accuracy: 0.8743
Poisoned Accuracy: 0.3841
Poisoned + DP Protection Accuracy: 0.7909
Drop (Clean - Poisoned): 0.4906
Drop (Clean - Init): 0.0819
Drop (Clean DP - Init): 0.0815
Drop (Poisoned - Init): -0.4087
Drop (Poisoned DP - Init): -0.0019
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8804
Clean Recall: 0.8747
Clean F1 Score: 0.8741

Clean DP Precision: 0.8802
Clean DP Recall: 0.8743
Clean DP F1 Score: 0.8737

Poisoned Precision: 0.5302
Poisoned Recall: 0.3841
Poisoned F1 Score: 0.4037

DP Protection Precision: 0.8030
DP Protection Recall: 0.7909
DP Protection F1 Score: 0.7893

Summary

FL Simulation Complete
Task: 2a731c7b-9d90-4ed8-a63d-cfa42b2e3bbf
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8747
Clean DP Accuracy: 0.8743
Poisoned Accuracy: 0.3841
Data Poison Protection Accuracy: 0.7909
Drop (Clean - Poisoned): 0.4906
Drop (Clean - Init): 0.0819
Drop (Clean DP - Init): 0.0815
Drop (Poisoned - Init): -0.4087
Drop (Poisoned_DP - Init): -0.0019
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8804
Clean Recall: 0.8747
Clean F1 Score: 0.8741
Clean DP Precision: 0.8802
Clean DP Recall: 0.8743
Clean DP F1 Score: 0.8737
Poisoned Precision: 0.5302
Poisoned Recall: 0.3841
Poisoned F1 Score: 0.4037
DP Protection Precision: 0.8030
DP Protection Recall: 0.7909
DP Protection F1 Score: 0.7893